

USER MANUAL

LVTS-512314-G4 LFP Battery



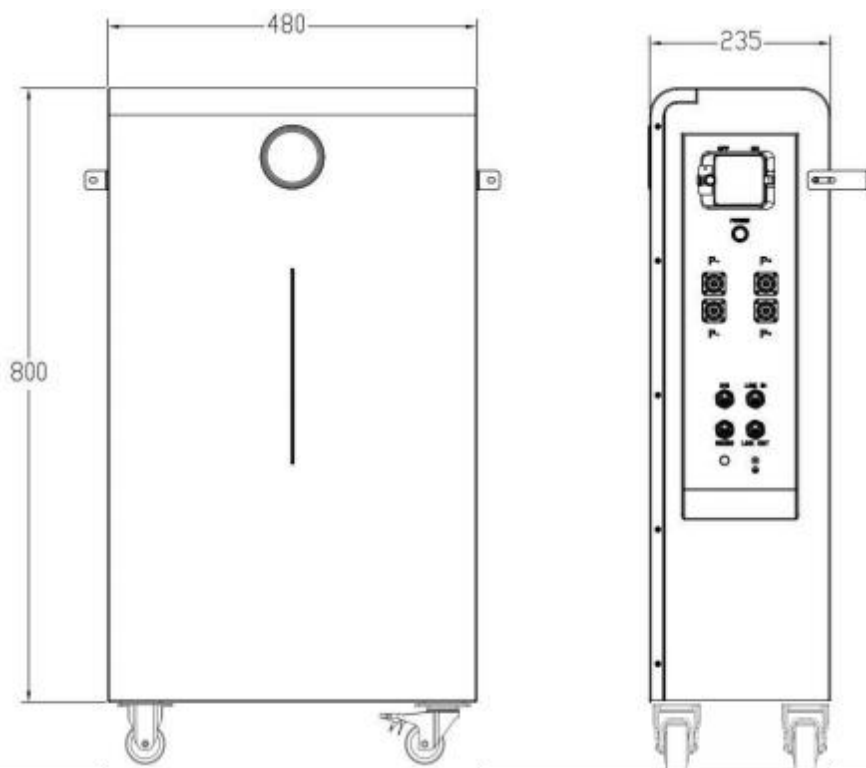
Contents

1 Product Overview	1
1.1 Appearance	1
1.2 Switch, LCD, LED & Communication Port	2
1.2.1 Color touchscreen	2
1.2.2 Communication Port Definition	6
2 Installation Guide	7
2.1 Checking Deliverables	7
2.2 Tools	8
2.3 Installation Instructions	9
2.3.1 Installation Step	9
2.4 APP installation and usage(Wifi function)	11
3 Technical Specifications	12
4 Maintenance	13
4.1 Recharge Requirements During Storage	13
4.2 Recharge Requirements When Over Discharged	13

1 Product Overview

LVTS-512314-G4 is a 51.2V314Ah Lithium battery that can be floor mounted. Note: LVTS-512314-G4 is NOT Suitable for life-sustaining medical devices.

1.1 Appearance





1.2 switch, LCD, LED & communication port

1.2.1 color touchscreen

- 1) swipe the screen to see "SN Info", "capacity Info", "Temp Info", "cell voltage", "current Event", "History Event", "set protocol" & "Display Time".
- 2) If current protocol is incorrect, tap "CAN" checkbox and select correct protocol.
- 3) If current time is incorrect, tap "set Time" button and change time then tap "save" button.





Temp Info

T1: 26.0°C T3: 25.8°C
T2: 25.7°C T4: 26.0°C
T_MOS: 27.8°C
T_ENV: 33.8°C

Cell Voltage

01:3299mV 09:3299mV
02:3300mV 10:3298mV
03:3299mV 11:3299mV
04:3298mV 12:3299mV
05:3299mV 13:3299mV
06:3298mV 14:3299mV
07:3298mV 15:3299mV
08:3299mV 16:3299mV

51.69V 0.00A

Current Event

Alarm :None
Protect :None
Fault :None



History Event

001 07.16 09:34 FUL ▲
pLSW p0VP
002 07.16 09:34 pLSW
003 07.16 09:34 sUVP
004 07.12 17:35 e0TP
005 07.11 17:03 e0TP ▼
006 07.11 14:38 e0TP
e0TP





1.2.2 communication port Definition

1.2.2.1 CAN/RS485 to PCS

port definitions	RJ45 pin	Function
	1	RS485-B
	2	RS485-A
	3	GND
	4	CAN-H
	5	CAN-L
	6	NC
	7	RS485-A
	8	RS485-B

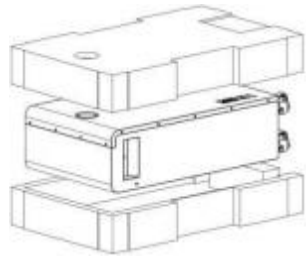
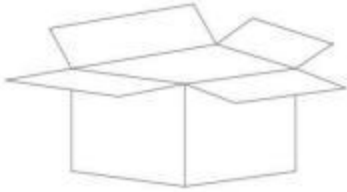
1.2.2.2 RS232 to PC

port definitions	RJ45 pin	Function
	1	RS232-TX
	2	GND
	3	RS232-RX
	4	NC
	5	NC
	6	NC
	7	NC
	8	NC



2 Installation Guide

2.1 checking Deliverables

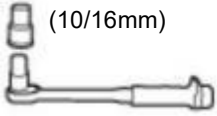


NO.	pictures	Quantity	Description
1		1 pcs	Battery
2		1 pair	Power cable
3		1 pcs	comm cable
4		1 pcs	Expansion Bolt
5		1 pcs	wall connector



6		1 pcs	Battery hanging bolt
---	---	-------	-------------------------

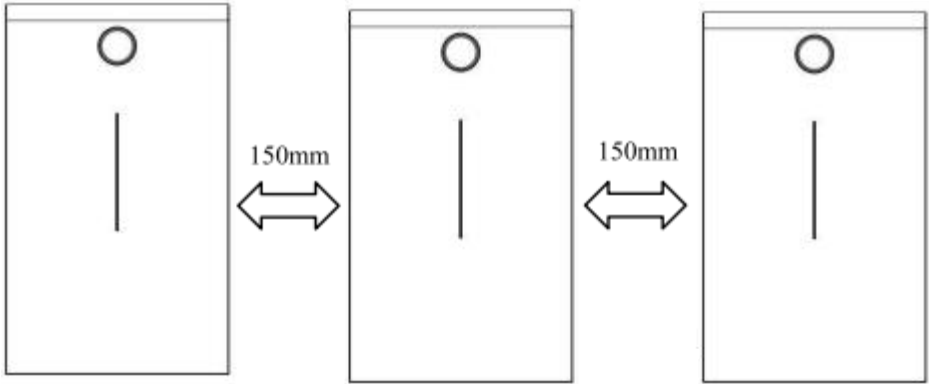
2.2 Tools

Tools			
Installation	knife 	Measuring tape 	socket wrench (10/16mm) 
	Hammer 	cross screwdriver 	Hammer drill 
protection	ESD gloves 	safety goggles 	safety shoes 



2.3 Installation Instructions

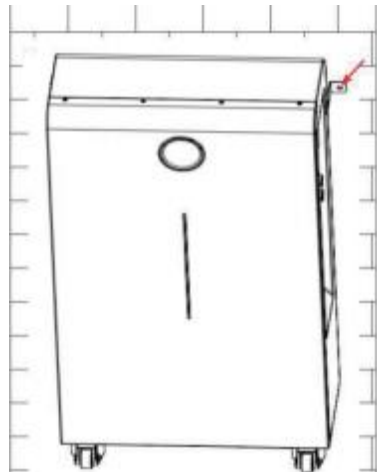
Minimum mounting distance requirement:



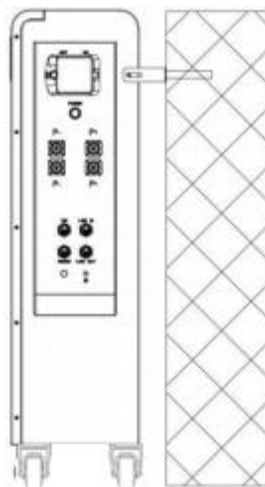
2.3.1 Installation step

step 1 place LVTS-512314 on the floor, close to wall.

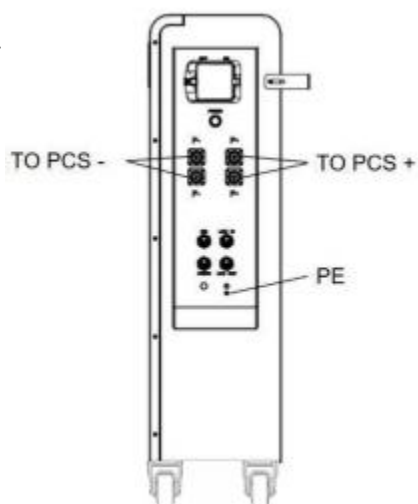
Determine hole position using marker.



step 2 drill hole on the wall,
fix wall connector using expansion bolt

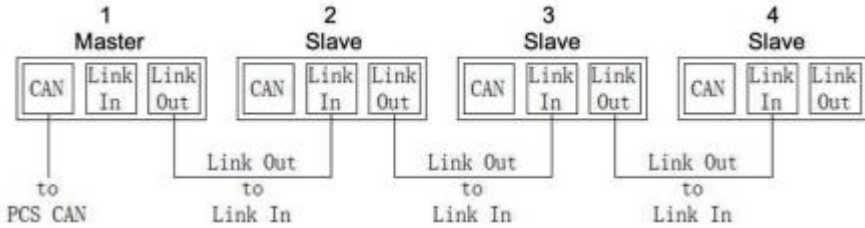


step3 connect pE Cable & power cable.

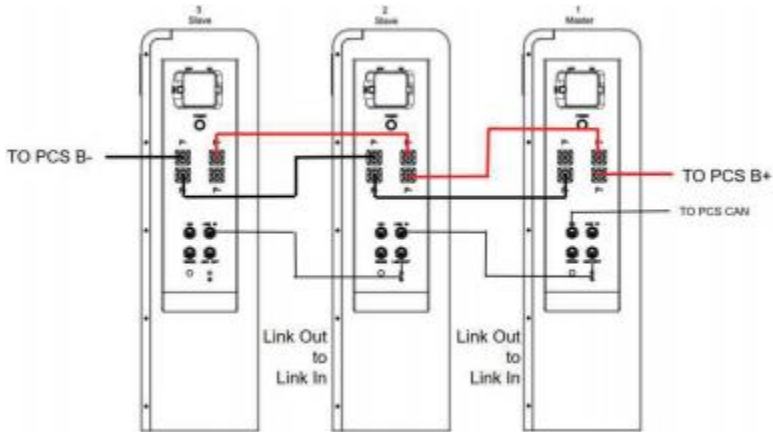




Step 4 Connect communication cable.



Step 5 Refer to the following diagram when multiple batteries are connected in parallel:



2.4 APP installation and usage(Wifi function)

You can obtain the mobile application and user manual that match the product by scanning the QR code displayed in the image.



3 Technical specifications

Basic project		parameter
Nominal voltage		51.2V
Nominal capacity		314Ah
Nominal Energy		16076.8wh
charge voltage		56.16V
charge current		150A
Discharge voltage Range		45.6V—56.16V
Discharge current		200A
communication Mode		CAN/RS485
working	charge	0°C ~45°C
Temperature	Discharge	-20°C ~60°C
	short Term	-10°C ~45°C
storage	(within 1 month)	
Temperature	Long Term	0°C ~35°C
	(within 1 year)	
storage Humidity		<95% RH
cell Type		LiFePO ₄ , Lithium Iron phosphate
size		H800*W480*D235(mm)
weight		117.5KG (wall Bracket Included)
IP Level		IP65

4 Maintenance

4.1 Recharge Requirements During stor'age

Batteries should be stored in temperature between - 10°C ~ +45°C , and recharged regularly according to the following table with 0.2c (60A) current to 50% Soc after longtime storage.

Recharge requirement during storage

storage Temperature	storage Relative Humidity	storage Time	SOC
Below - 10°C	/	Not Allowed	/
- 10~0°C	5%~70%	≤1 months	30%≤SOC≤60%
0~25°C	5%~70%	≤12 months	30%≤SOC≤60%
25~35°C	5%~70%	≤6 months	30%≤SOC≤60%
35~45°C	5%~70%	≤1 months	30%≤SOC≤60%
Above 45°C	/	Not Allowed	/

4.2 Recharge Requirements when over Discharged

please recharge over discharged (90% DOD) batteries according to the following table, otherwise over discharged battery will be damaged.

Recharge requirement when battery is over discharged

storage Temperature	storage Time	Note
- 10~25°C	≤15 days	Battery disconnected from PCS
25~45°C	≤7 days	
- 10~45°C	< 12 hours	Battery connected to Pcs

Attention: Disposal of batteries should follow local regulations.